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The False Science in the Biomedicine - a Dilemma

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Background: Today we live in a “scientific age” and that most people believe all the information that is presented to them through the media, through the tricks of various industries, especially the pharmaceutical industry, and all because of a misunderstanding of statistical data and a bad interpretation of evidence, which ordinary people actually cannot understand, because of the way it is presented to them. **Objective:** The aims of this study were: to present false science through homeopathy, placebo effect, nutrition and antioxidants and explain the use and abuse of drugs through false science, the bad side of pharmacy and show the roles of the pharmaceutical industry on the important examples in the past. **Methods:** Author used a descriptive method of analysis from the appropriate scientific literature. **Results and Discussion:** The meta-analysis of the use of adequate sources about facts described in the article helped the author to recognize nonsense and intellectual fraud. The analysis is based on exposing pharmaceutical companies, nutritionists, various statisticians, but also the media that publish scientific articles about non-existent experiments and abuse science and present false facts. Pharmaceutical companies, like all other commercial enterprises, are profit-driven in their practice. Therefore, they do not produce unprofitable medications, but expensive and chargeable ones, even if they are useless, at the current level of expansion, further growth in profits requires the maintenance of existing medications and the creation of new diseases. A naïve view of science opposes any opinion about science that comes from outside science without ever asking what science is, how it differs from other forms of cognitive and spiritual questioning, what foundations it is built on, what is a scientific revolution, what is truth and what are the standards of truth in science, is there progress in scientific knowledge, what are the possibilities and limits of science in explaining and understanding reality, what is its ultimate meaning, and finally what are real information and what are misinformation. **Conclusion:** Regardless of all the arguments there will still be both paths along which human knowledge moves; in addition to the scientific, its adherents will also have an esoteric understanding of the world. Science is something like “never ending story” (by Masic) and will not stop from advancing - it will continue to bear fruit that will be used and abused.

Keywords: science, pseudoscience, bad science, pharmacy, COVID-19 vaccines, misinformation.

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1. BACKGROUND

True knowledge is gained through scientific research. The highest degree of knowledge is the ability to investigate scientific problems (1). The best way to get to the truth is the knowledge on scientific methods (2, 3). Well-defined rules in all stages of research consist of ethical code in good scientific practice (GSP) (1). Finding truth in science is not only ensured by consistent application of the scientific methods. Honesty of scientist who searches for truth is of crucial importance (2). The modern science is multidisciplinary and multi-professional, so that the work resulting from this cooperation is usually signed by several authors. Among the first examples of the evaluation process as such is its initiator, Henry Oldenburg, who in 1665, founded and edited the journal “Philosophical Transaction of the Royal Society” in London, as the earliest scientific publication of its kind in English. In the early 9th century, it is possible on the basis of available data to discover the practice of certain authors, the Arab philosopher Al-Gazaly used reviewing method for

quality assessment of written text. Yusuf al-Kindi (cca. 800-870), for example, shared his written work with his colleagues to give him their critical opinion on the work. On the other hand, in our time, according to the Association of American Historians, professional review is described as: a) Verification means that the book, monograph or paper, or research proposal, is read and evaluated in the field of the topic, and the language and documents of the author, by experts in a certain period of time. As prominent experts in the field of knowledge that the author deals with, reviewers prepare an analysis for a committee made up of experts of scientific importance for the article with questions; b) Does the author demonstrate knowledge of current developments in this field? c) Are the research procedures, processes, and methodologies, in accordance with professional standards? d) Does the author offer original arguments and provide valid facts that support his work? If the relevant claims are weak or not in the presented article, the reviewers suggest a revision that will correct the article and re-evaluate

it before funding the project or before accepting it for the journal. What this text is important to define real science and false science or pseudoscience?

1.1. What is science?

Science (in Latin is *scientia*, in Greek is *λογος* – *logos*, in Arabic is *ilm*, in English is knowledge, science, learning) is a system of arranged and systematized knowledge about us and the (material and immaterial) world that surrounds us (1). It is a large and comprehensive group of information about a subject as well. The word science was specifically used for information about the physical universe. As knowledge increased, certain methods proved more reliable than others, and today scientific methods are the standard for science. These include the use of careful observation, experiments, measurements, mathematics and repetition (1).

According to standard dictionaries, science is a systematic endeavor that creates and organizes knowledge in the form of explanations and theoretical predictions about the Universe (space) (1). Modern science is discovery as well as invention. It is the discovery that nature in general operates in a sufficiently stable manner to be described by laws, even mathematical ones. Scientific research requires technical design, abstractions, apparatus and organization for the presentation of results that resembles legal descriptions. In older and closely related meanings, science also refers to such a body of knowledge of itself that can be rationally explained and reliably applied. The material and spiritual progress of humanity is primarily conditioned by the development of science. Based on this, it can be said that the scientific vocation is one of the most demanding vocations that a person can choose. Science is the totality of acquired knowledge expressed in a systematic form, with the aim of studying a certain field of phenomena, type of phenomenon or area of existence, to the end and completely and to acquire such knowledge about them, which will have lasting value and which can be the basis of human practice (1). Science is not a special kind of cognitive power. It is born from the same kind of knowledge and the same cognitive power, which is innate to man and with which all normal people have at their disposal. Science is an organized, systematic, repeatedly repeated observation of phenomena, whether they are phenomena of nature or phenomena of history or the human spirit. The scientific vocation seeks the whole man, occupies its thoughts, and drain its physical and spiritual strength.

At wikipedia is written: "science is a systematic endeavor that builds and organizes knowledge in the form of testable explanations and predictions about the universe" (1), and the earliest written records in the history of science come from Ancient Egypt and Mesopotamia in around 3000 to 1200 BCE, besides historians stated that science may be as old as the human species (1). Their contributions to mathematics, astronomy, and medicine entered and shaped Greek natural philosophy of classical antiquity, whereby formal attempts were made to provide explanations of events in the physical world

based on natural causes (1). During the Muslim the Islamic Golden Age (1) and later by the efforts of Byzantine Greek scholars who brought Greek manuscripts from the dying Byzantine Empire to Western Europe in the Renaissance. The recovery and assimilation of Greek, Roman and Islamic works (9th to 15th century) into Western Europe revived "natural philosophy", (1) which was later transformed by the Scientific Revolution (Humanism and Renaissance) as new ideas and discoveries departed from previous civilisations and the scientific methods soon played a greater role in knowledge creation. Modern science developed in the last century when many of the institutional and professional features of science began to take shape along with the changing of "natural philosophy" to "natural science" (1).

Modern science is typically divided into three major branches: natural sciences (e.g., biology, chemistry, and physics), the social sciences (e.g., economics, psychology, and sociology), and the formal sciences (e.g., logic, mathematics, and theoretical computer science). Applied sciences are disciplines that use scientific knowledge for practical purposes, such as in engineering and medicine, or better to say - biomedicine (medicine, stomatology, veterinary and pharmacy) (1). In order to be considered from a scientific point of view, the object of observation must be subjected to repeated testing by independent observers. Using scientific methods to make new discoveries is called scientific research, and the people who do this research are called scientists. In a narrower sense, science is called experimental, while the purpose of applied science, for example, engineering, is the practical use of scientific knowledge. Scientific hypotheses are educated proposals to explain phenomena or reasonable assumptions about the nature of the universe. New knowledge in science is advanced by research from scientists who are motivated by curiosity about the world and a desire to solve problems (1). Contemporary scientific research is highly collaborative and is usually done by teams in academic and research institutions, government agencies, and companies. "The practical impact of their work has led to the emergence of science policies that seek to influence the scientific enterprise by prioritizing the ethical and moral development of commercial products, armaments, health care, public infrastructure, and environmental protection" (1).

A professional in the field of science is known as a scientist. The foundation of science rests on the great intellectual power that distinguishes man the most from all living beings. Thanks to his intellectual power, man has an overall picture of the world, which it constantly explores and expands, perfecting that picture. Thinking is the highest psychic process and activity that occurs in the world. It consists in man's ability to create general ideas or concepts about everything that comes within the reach of thought and becomes its object (1)..

A scientific theory is a hypothesis that has been confirmed by repeated observations and measurements. Scientific theories are usually given as mathematical forms

and are always subject to rejection if future experiments disprove them. In the modern world, scientific research is the most important activity of all developed countries, and scientists are expected to publish their findings in reference journals, scientific periodicals where referees check the facts of the article, before it is published. Even after publication, new scientific ideas are not generally accepted until the work is replicated. Scientific literacy is the ability of the general population to understand basic concepts related to science (2).

1.2. Goals of the science

The goal of science is to know the laws according to which something happens in nature and society (1). Knowing the law makes it possible to predict future events. The starting point is noticing the problem. The problem should be defined clearly, in all essential elements without redundant and unnecessary facts (which can only obscure the problem itself). This is followed by the collection of relevant facts and the establishment of possible solutions. Those possible solutions are assumptions, or hypotheses. A hypothesis provides an explanation of the investigated phenomenon, but that explanation is only more or less satisfactory (there is logic in it); it has yet to be proven. Not every statement can be considered a hypothesis. A good hypothesis should be clearly related to the facts on which it is based to explain all the known facts to which it relates. That it is more economical and elegant than other hypotheses, and this will be if it requires fewer additional assumptions in accordance with another accepted scientific knowledge; and to be more useful than other (as "rival" hypotheses in explaining the given phenomenon). That unknown facts can be predicted based on it. When such a hypothesis is built, it goes to the test, it is tested by different researchers in order to avoid the subjective assessment or "blindness" of an individual researcher. A hypothesis that satisfies the above conditions becomes a scientific theory. A theory carries more weight, is more convincing than a hypothesis, and can be considered highly probable. But the theory is not yet fully proven, it is only very likely. And theories need to be proven and they are constantly being proven. A proof is a claim based on direct experience and on the basis of other already proven claims. A proven statement leaves no doubt because the given phenomenon cannot be explained by any other statement. When a theory or a law is reached, the scientist's job is almost over. Almost, because the experience is bitter. It happened that what was considered proven for a long time, even for centuries, was overturned. That is why scientific claims are constantly being tested in a constant attempt to be refuted, intellectually or empirically. It is enough only once, only in one case, to disprove a claim and it falls (5).

1.3. What is pseudoscience

We live in a world where goods are sold not only by quality, but also by marketing. In order to overcome the defense that potential buyers create against such manipulation with their critical attitude, many sellers will offer

their goods with false information about their origin and effectiveness. In a way, knowledge is also a commodity that has its price. A wide range of "intellectual property" is offered on the knowledge market; as everywhere, here too there is a lot of quality offer, but also scrap. Any reasonable buyer will look in this market for knowledge that is verified by science or originates from its institutions.

"Pseudoscience is when people put on a scientific cloak to make themselves convincing on a superficial level. Tearing off that cloak is always fun". Ben Goldacre. Pseudoscience arose as a result of an effort to present some views and theories as scientific, even though they are not. A pseudoscientific theory is often not even testable because it is so broadly formulated that it is consistent with every possible course of events and nothing could disprove it. In imitating science, pseudoscientists use all means to disguise their performance as scientific: jargon, institutions and even the titles that scientists normally use. Pseudoscientists, in fact, start from the desired conclusion and then move backwards and devise a procedure to confirm that conclusion.

Just as scientists reject a theory that is not confirmed by experiments, so pseudoscientists reject the results of experiments that do not support their favorite theory. A good illustration for this is the words of Sherlock Holmes, the popular detective created by Sir Arthur Conan Doyle: *"It is a big mistake to draw conclusions before we gather the facts. Instead of twisting the facts to fit a theory, we should change the theory until it fits the facts."*

If the procedure is turned "upside down", its result is not the truth but manipulation, behind which is usually hidden the desire for easy profit or some other goal. This reverse procedure sometimes even turns into a loop where the theory does not rely on facts but "holds" itself in the air. Pseudoscientists give various excuses to explain why science does not accept the results of their work. The loudest are the supporters of the conspiracy theory, who often express the belief that there is an organized conspiracy against them among all the world's scientists. If this interpretation is to be believed, the scientific world considers itself threatened by all those who do not think the same as them, so it "closes ranks" in the fight against competition. This argument, however, is untenable because there are many scientific institutions in the world that are not only mutually independent, but consider themselves competitors. That's why each of them will be happy to accept and develop anyone's reasonable idea that has not already failed the test and that offers at least some hope that it will lead to some useful goal. It should be said that there are a lot of pseudo-scientific theories that are in line with the facts that are presented along with them, but the facts are so chosen that they mostly prove nothing. Just as a selection of experimental results can produce false evidence for any theory, so a selection of facts can support an incorrect claim.

There are two types of pseudoscience. One is relatively harmless and is used by people who have no scientific education and have never been involved in science. This

species is relatively easy to recognize, especially since its devotees use the language of science unskillfully. In doing so, they go to one of two possible extremes: either the jargon is adapted so that the general public can easily understand it (which is mostly seen in advertisements offering various esoteric means under the guise of scientific achievements) or it is artificially complicated and “enriched” in such incomprehensible terms that it does not carry any useful information but aims to impress the listener and convince him that he is dealing with an excellent expert. What both extremes have in common is extremely unskilled improvisation and handling of scientific language and data.

Another type of pseudoscience is represented by people who are engaged in science, who are scientifically educated but never managed to grow a mystical understanding of the world. In an attempt to grasp something that, in their opinion, is above science, they are often not even aware of how much they are degrading their scientific work (6). “Bad science is a science in which the nonsense served by nutritionists, homeopaths, but also cosmetic and pharmaceutical companies is systematically exposed. Namely, on the one hand, it is said how people misuse science, and on the other hand, how real science works” (stated British physician Ben Goldacre in his book “Bad Science” published in 2009) (7). False (Bad) science is not the same as pseudoscience, but it is very close to pseudoscience in terms of research methods.

2. OBJECTIVE

The aims of this study were to resent false science through 3 different points (aspects: a) to explain homeopathy, placebo effect, through nutrition and antioxidants, the use and abuse of drugs, as the bad side of pharmacy; b) to describe domination of medical information and information and communication technologies (ICTs) in diagnostics and therapy treatments for healthcare protection; c) to show the roles of the pharmaceutical industry on the examples of drug products and vaccines.

3. MATERIAL AND METHODS

This paper was written using retrospective and descriptive methods of analysis the facts from the available literature. The analysis of available literature included professional books, scientific articles and other literature available on online browsers. A book written by the author Ben Goldaker called “Bad Science” (published in 2009) (7) actually describes in detail the method and way to recognize nonsense and intellectual fraud. The book is based on exposing pharmaceutical companies, nutritionists, various statisticians, but also the media that publish scientific articles about non-existent experiments and abuse science and present false facts. There are many relevant good articles stored in social databases on Internet browsers, which talk about pseudoscience and science, but also about the basic difference between pseudoscience and science, and how to recognize

bad science today. The point of writing the paper is the presentation of bad science, through some of the terms that are covered in the book “Bad Science” and some other monographs with an emphasis on the use and abuse of drugs through bad science, but also on the general presentation of bad pharmacy and the pharmaceutical industry through the examples mentioned.

4. RESULTS

4.1. The basic significances of the Arabic-Islamic medicine in the middle age

Medicine and Pharmacy are among the oldest human activities (3-6). The history of these sciences is closely connected with the socio-economic, cultural and religious circumstances of the nations within which they developed (3). A significant cultural transformation and revival of medicine and pharmacy for the Arabs was brought about by the appearance of Mohamed Profet (Muhammad s.a.w.s.). The Qur'an as God's revelation and as a guide for all mankind (3). In the month of Ramadan, the year 610, Muhammad a.s., received the first words of God, the imperative Iqre-learn, read, study. It was this first revelation that later inspired thousands of scholars around the world (2). The appearance of God's last revelation gave a strong impetus to the development of health opportunities, first in the Muslim world, and then spread throughout Asia and Europe. Islam as a religion that propagates science as a path of truth has given a great impetus to the development of science, including pharmacy and medicine. Islam is not only a religion but also a philosophy of life, with a series of completely defined and mandatory regulations on hygiene and cleanliness, ritual washing before prayer, fasting, ban on alcohol and everything intoxicating, ban on pork consumption, circumcision as a prevention of genital cancer, and prayer five times a day accompanied by an element of exercise (3). In addition, the Qur'an prescribes Islamic marriage, which hinders the spread of sexually transmitted diseases and the development of prostitution (3). In Gonde-shapur, in western Persia, long before the founding of Baghdad and its medical school and hospital, there was a large hospital and a famous medical school. In that school, Greek medical science was nurtured and famous Greek and Roman works were translated. When the need to educate medical scientists was felt, after the unusually rapid expansion of the new Arab state and the great flourishing of Arab culture, progressive Arab rulers of the Abbasid dynasty invited scientists from various quarters to the newly established capital Baghdad. During the 150 years (from 750 to 900 AD) in this new cultural hotspot, all the most famous works of Greek and Roman medical writers were translated into Arabic (3). The works of Hippocrates, Galen and Dioscorides were especially fond of being translated. Young Islamic pharmacy and medicine had at its disposal all scientific Greco-Roman literature and with Persian and Indian works. Islamic culture, which made a huge contribution to the development of biomedical sciences, especially pharmacy and medi-



Figure 1. Classics of Arabic medicine: ar-Razi, ibn al-Haitam, az-Zahravi, Ibn Sina, al-Biruni and ibn Rushd

cine, reached its greatest stage of development between the 9th and 11th centuries. After the fall of the Roman Empire in the 5th century and later in the 7th century (Hijra -622), the spread of knowledge about medicine in the Islamic Caliphate began. In that period, there was a „School of Commentators on Greek Works“, and Muslims are credited with preserving the knowledge of ancient physicians (3). In the Renaissance, Arabic translations introduced Europe to ancient medicine. The founding of the first modern hospitals, pharmacies, medical schools, city medical services dates back to the rise of Arab medicine and culture. Islamic scientists have also improved and perfected pharmacology and chemistry. At the beginning of the caliphate, the Islamic people formed a great empire with strong trade ties, which gave the pharmacy a huge boost. Muslims began to use new medicinal plants coming from India or Persia. The study of foreign drugs, the research of chemical relations between drugs, led them to formulate principles that will serve as a basis for pharmacy. The world's first pharmacy was founded by Caliph al-Mansur in Baghdad in 754, and from there the establishment of pharmacies slowly spread throughout Spain and Italy, and later throughout Europe (4). For 150 years (750-900) in Baghdad, as a new cultural center, almost all the most famous works of Greek and Roman medical writers were translated into Arabic. Among the translated works we have the first books on the medicinal herbs of the Greek THEOPHRAST and the first books on the medicines of DIOSCRIDIS under the title "De materia medica" contained in five volumes. Dioscridis' "De materia medica" includes a comprehensive overview of ancient pharmacy and medicine and is the best source of pharmaceutical knowledge to date. It is the first pharmacopoeia and builds on the experience of Islamic-Arabic science in which form will dominate for four centuries (3).

The advanced rulers of the Arab states founded translation schools and helped cooperate with Christian scientists in the field of medical sciences. Such scientific progress leads to the formation of golden Arab medicine, which greatly contributes to the development of pharmacy. In the field of biomedical sciences, Muslims built upon the early legalities of the Indians, Syrians, Romans, and most of all the Greeks. Medicine was defined by famous Muslim physicians such as ar-Razi and Ibn Sina. The Arabs improved chemistry and pharmacology, and good trade relations and the availability of new plant species resulted in the enrichment of Pharmacognosy.

It is important to note that we find the first pharmacies among the Arabs and that the legal regulations related to the operation and management of pharmacies have been improved. The Qur'an, as the moral code of Muslims, led to the improvement of professional ethics. In addition to all the advantages provided by Islamic medicine, as a form of alternative medicine, it should be emphasized that alternative medicine is not, nor can it be, a substitute for classical medicine. Modern medicine is a basic and irreplaceable system of health care throughout the world that is continuously improved and uses the most modern diagnostic and therapeutic methods and equipment. Within modern medicine, modern technology gives irreplaceable results in all its segments. Due to the above facts, classical medicine deserves full respect and dedication to treatment. In any case, alternative medicine and classical medicine should not ignore each other, but in order to provide better health care, they should be used together for the benefit of the patient.

Pharmacology has reached its peak in Spain. The Arabs used the mineral and plant wealth of Spain and brought many useful plants to the area. It is an extremely important fact that the Arabs, with their contribution, led to the gradual differentiation of medicine and pharmacy. They founded the first hospitals with a permanent medical service and a hospital pharmacy. Therefore, the emergence of pharmacy, as an independent profession and science, means that the social forces and cultural heritage of the people within which it developed, have reached such a level of development, that they have enabled this distinction (4).

The scientific research method that will be used is the compilation method, i.e. the procedure of taking over other people's results of scientific research work, observations, attitudes, conclusions and cognitions, with a personal review of the taken over parts of scientific research work. Since the original experimental scientific research is not included, this thesis has the scope of a retrospective study. In modern times, it is noticeable that the number of patients with known or new diseases is constantly increasing, which increases the need for their treatment. It often happens that the medicine does not work in accordance with expectations or that people fall into pathological conditions that have no confirmation in classical medicine. Basic chemical processes including distillation, crystallization and sublimation were also discovered. Muslim pharmacists and physicians showed a special interest in natural products and



Figure 2. Top ten most influential scientists

medicines because they believed that natural medicines were a sign of God's generous giving to man, which motivated Muslim pharmacists to search for medicines in nature. Islamic culture, which made a huge contribution to the development of biomedical sciences, especially pharmacy and medicine, reached its greatest stage of development between the 9th and 11th centuries, and brought a significant number of new discoveries during the 12th, 13th and 14th centuries. From the 12th century until the Renaissance period, Arabic medical manuscripts were translated into Latin and became available to the countries of the Iberian and Apennine peninsulas. The most significant contribution of Arab physicians, pharmacists and philosophers is reflected in the fact that by translating the best medical and philosophical works of previous civilizations they preserved their medical and cultural heritage, and then supplemented it with their own experiences during the Dark Ages. progress.

The development of pharmacy was indisputably contributed to by Islamic-Arabic medicine, which significantly enriched pharmacology with a series of new drugs through a series of encyclopedic works and a series of theoretical and practical discussions from almost all medical disciplines. Many tinctures, syrups, ointments, then alcohol and distilled water, and many other pharmaceutical preparations have come into use thanks to Islamic-Arabic medicine. The arrival of Islam has led to the revival of science in all fields, and it is especially necessary to emphasize the contribution of Islam to the development of public health. The need for ritual washing gave birth to the cult of water in the Islamic world. This religious duty led to the construction of aqueducts, waterworks, public baths, hammams, fountains. It is important for the history of pharmacy that we find the first beginnings of pharmacy as an independent profession among the Arabs. The Arabs developed „folk medicine“, which was transmitted orally from generation to generation. Even before Islam, the Arabs came into contact with people who had developed medical literature, medical schools and hospitals in which doctors were professionally educated. Greek medical schools and centers of science, such as Kos and Alexandria, had a great influence (3). Knowledge of medicine spread through the Islamic Caliphate between the fall of the Roman Empire in the 5th century and the European Renaissance in the 15th century. However, the rise of Islamic medical institutions began during the 9th century and coincides with the golden age of the Abbasid Caliphate in the east (749-1258) (5). to the Eastern Roman Empire, to Byzantium.

The Abbasid Caliph Ma'mun took a giant step to-

wards establishing translation departments and medical schools when he established Dar el-Hikma (Home of Healing) in Baghdad. This main institution contained a translation school run by the physician and philosopher Hunayn ibn Ishaq al-Ibadi (810-873), who was also the chief translator at the time, and happened to be the son of a pharmacist from the city of Hire in Iraq. These medical schools established a base of medical practice at the time and contributed greatly to innovations in medical practice, outpatient patient care, and mobile clinics (3). Among these schools and hospitals, the largest and most famous was the El-Nuri Hospital, in Damascus (1160), which remained active for almost three hundred years, and the Mensuri Hospital in Cairo, Egypt (1276). At one time, Baghdad had close to 600, while Cordoba had more than 500 hospitals in Spain. The largest hospitals had libraries, outpatient clinics and medical schools. These hospitals have done careful research into the separation of patients with fever or infectious diseases, as well as the mentally ill. Medical education was transferred from these hospitals. Students sought theoretical and practical education there, and well-known doctors and surgeons were chosen to serve in these hospitals. Mensuri Hospital was the first hospital to nurture science, learning and social care. She divided the wards into women, children and those who are recovering, and wards suffering from specific diseases. In addition, there were smaller libraries and private collections, all of which contained no less than 100,000 books. These libraries contained medical works, among other scientific works such as those in chemistry, astronomy, geometry, philosophy and other branches. At that time, the eastern and western capitals of the caliphate became centers of civilization, and medical institutions and scientific research were funded by the state.

Islamic physicians have emphasized both clinical and basic medicine in their lectures. Medical students were required to be educated in the basic sciences and to have appropriate knowledge from the works of medical authorities such as Galen, Hippocrates and El-Zahravi, among others (3). Medical knowledge was compiled into a collection of manuscripts so that clinical tests could be reasoned to a certain extent and students were examined in the basic sciences. Islamic scientists have also improved and perfected pharmacology and chemistry. They described many new medicines such as hay, camphor, nutmeg, cloves, etc., and used new solvents such as rose water and orange water. They also used aldehydes, alcohol and other solvents as well as perfected methods of testing the purity of metals and chemicals. Their per-

sistence in searching for methods of converting metals to gold has resulted in the discovery of several chemicals such as acid metals, bismuth, ammonia and mercury compounds (4). Arabic words such as alcohol (ar. kuhul), syrup (ar. surub) and others, are widely used. Basic chemical processes including distillation, crystallization and sublimation were also discovered. It is also known that Islamic physicians used cannabis sativa indica (ar. Kunnab hindi; hashishatul kaif) for inhalation as an anesthetic. El-Kindi's work on the method of prescribing and precisely determined dosage of drugs is well known. He applied the law of geometric progression in prescribing drugs. And finally, the word about Muhammad ibn Abdullah al-Ash'at from Mosul in Iraq as an expert in medicine and pharmacology brought students from far and near cities to hear his lectures. This sea of knowledge, which lasted for almost four centuries, was later stopped by the Mongol invasion of the Eastern Caliphate and the decline of the Western Caliphate in Spain (5, 6).

4.2. Bad science through homeopathy and placebo effect

Alternative medicine methods are very popular in the West. Although some conducted scientific studies have proven that homeopathic remedies are no more effective than a simple placebo, people still happily buy homeopathic preparations. Because they pay huge amounts of money for it, homeopathy has become a billion-dollar business. Scientific research has not shown that homeopathic preparations are effective, and the assumptions on which their supposed effect is based are unreliable. The scientific community considers homeopathy a fraud and quackery, and homeopathic preparations unethical (8). Using the example of homeopathy, or the method of alternative medicine, Boldvek himself explained that the role of medicine, and people's beliefs about the body itself and treatment, as well as evidence about the good and bad sides of treatment methods are important. One of the most important questions in science is how to know if a certain method of treatment is effective (7).

Homeopathy is a method of alternative medicine whose rule is "like cures like" (lat. *Similia similibus curentur*), and its foundation was laid by the German physician Samuel Hahnemann (1755-1843). (Hahnemann's rule is an ipse dixit axiom, or a never proven assumption on which he founded homeopathy). His thesis that *homeopathic preparations work better the more diluted they are* is in complete contrast to pharmacological studies which indicate that *a higher concentration of a solute must have a greater effect than a lower concentration*. The model was developed and published in 1796 in the book *Materia Medica* his conclusions, which he reached by translating the work on malaria. When he read that quinine is a cure for malaria because it is bitter, he imagined it because he knew other bitter substances, which have no therapeutic value. To better understand the effects of quinine, he decided to try it on himself and found that he was developing symptoms similar to the

malaria he was treating. Then Hahnemann started experimenting with other previously known drugs in alphabetical order. He got about halfway and then laid the foundations of homeopathy (8).

Boldacre states that Hahnemann's theories differed from those of his competitors because he concluded—for lack of a better term—that a substance that could cause symptoms of a disease in a healthy person could be used to treat the same symptoms in a sick person (7). Homeopathic preparations have long been the subject of numerous clinical trials. Experiments have shown that at best homeopathy has a placebo-like effect, while at worst it can be harmful. Although some trials initially indicated positive results, systematic reviews determined that these were due to chance results, measurement errors, or examiner bias. The proposed mechanisms of homeopathy, which claim that substances in water leave a "signature" even when the water is so diluted that there is not a single molecule of the "active substance" in it, are in conflict with the fundamental laws of physics. For this reason, patients who decide to use homeopathy instead of tested medical methods risk setting an accurate and timely diagnosis and effective treatment of potentially serious conditions.

The most common criticisms directed at homeopathy are that it is a pseudoscience born of vitalism and that it resembles alchemy, and that patients fall under the placebo effect. Critics claim that homeopathy is illogical because the extremely small doses it uses cannot work. Illustrative example is that the concentration of 12C corresponds to the concentration of less than one gram of salt (0.36 mL) in a volume of water the size of the Atlantic Ocean (3.55×10^{20} L). For comparison, homeopathic doses are sold in bottles of 10 milliliters, which means that the patient has to drink 2.5 million bottles before drinking one molecule of the active substance.

Claims of homeopaths, that water, other solvents and inert carriers remember the properties of the substance that was in them, are not factually supported, and are in direct collision with the foundations of modern chemistry and physics. If substances left their "signature" with solvents, life could not exist. The basis of life is very complex and sensitive systems of biochemical reactions in water, and if water took on the properties of all the substances it has ever dissolved, these systems would never have developed. Hahnemann's original experiment with quinine, which is the starting point of all homeopathy, was never repeated, since quinine does not cause the described reactions. It is likely that they were caused by Hahnemann's allergy to quinine (8).

The literature on alternative therapy is certainly riddled with incompetence, but the flaws observed in the experiments are actually very common in medicine. It would be fair to say that all research has some flaws, simply because every experiment is a compromise between what would be ideal and what is practical or cheap. It is important that the research is always published with the stated methods and the results of the work, so that it

can be verified. The people themselves must be able to conclude whether there were any flaws in the research itself and to judge whether the presented, actually presented results are reliable.

The main criticism among conventional medical practitioners is that there is no active substance in a homeopathic medicine (as in all conventional medicines), and therefore it does not work. However, there are numerous studies on the effectiveness of homeopathic remedies and hundreds of well-documented studies on their effectiveness. Homeopathic medicine has been practiced for 200 years in almost the same way. At the same time, in that period, conventional medicine experienced hundreds of changes. Each time it was considered to be the right way of treatment, until after a short time it was found to be completely wrong, and sometimes very dangerous for the patient. Even today, we are witnessing the use of certain drugs that are later found to have caused more harm than good to patients.

In the latest research by the French Nobel laureate Luc Montagnier, it was scientifically proven that in the process of dilution and shaking in the solvent, the same electromagnetic signal remains as the one created by the same undiluted substance. The researchers believe that this effect is due to a change in the aqueous solvent at the level of its nanostructure. This study makes a significant contribution to evidence-based homeopathy (9).

In his book, Goldacre also mentions the placebo effect as an example of another deception in the field of medical research. The word "Placebo" means "to please, to make happy" in Latin. The term "*Placebo Domino*" is also found in the Bible, which translates as "please the Lord". Interestingly, the term Placebo was also used in the 14th century, referring to false mourners, who would falsely mourn and mourn the deceased during a funeral, in order to receive food and drink in return. Only in the second half of the 18th century did the word "Placebo" appear in the New Medical Dictionary, describing marginal practices that aim to "please" the patient rather than cure it. Today, the concept of placebo is already well established in medicine and is a way to observe the effect on the subject (patient) when he is given a fake medicine (without any effect on the body), without knowing it. It is also used when testing the effect of new drugs, then the test is carried out on several patients (experimental group), one group of subjects is given real drugs, and the other group (control) drugs in the same form as the real ones, but with completely harmless and useless content, which is not known to the respondents. After a certain time, the effects of real and fake drugs are compared. Placebo has unfortunately become excessively often used by pseudoscience in the sense that it explains the action of all kinds of alternative therapies, also various charlatans and quacks actively spread misinformation about the Placebo effect itself as something mystical and inexplicable, often associated with the power of "positive thinking" which takes advantage of the body's self-healing mechanisms. For this reason, below we publish

the text of one of the domestic scientists and researchers in this field, Miloš Babić from the University of Arizona, who once again proves that the truth is a little more complicated than it seems at first glance.

For example, needle treatment (acupuncture)—several small studies of acupuncture have shown that it significantly helps against certain types of joint and muscle pain. Hence, some have wondered if there might be some truth to the old Chinese story of chi energy flowing through the body through certain lines and which can become the cause of disease if the proper flow of energy is interrupted. To check this, several larger studies were organized. Patients were randomly divided into two groups, and sent to see two different groups of acupuncturists. One group consisted of "real" acupuncturists, masters of traditional Chinese medicine; they poked needles into the chi meridians, according to the ancient rules of this method of treatment. The second group consisted of "fake" acupuncturists, actors who poked patients with needles based on the principle of randomness. And what happened? Patients reported a great reduction in symptoms. Acupuncture helped them a lot... but equally for both groups. Therefore, pricking with needles is enough to help a person; knowledge of Chinese medicine and knowledge of qi meridian is not necessary at all. But is needling really so healing? In another study (which tested whether acupuncture helped women suffering from menopausal symptoms), scientists sent a group of patients to a fake acupuncturist who not only deliberately chose the wrong place to prick, but also used fake needles. A specially made needle would touch the skin but never sting, and was held in place only by a small piece of duct tape. The result? Many patients experienced significantly reduced symptoms, just like those who went to a "real" acupuncturist. The mere fact that they went to a therapy in which they had confidence was enough for that therapy to be effective (10).

The mechanism of action of the placebo effect is based on psychological action, because if a person believes that a certain drug (or therapy) helps, then the cerebral cortex affects the glands with internal secretion. Acting on glands with internal secretion (hypothalamus, pituitary gland), the subconscious affects the increase of the body's defense functions. With such action, the placebo effect affects autosuggestion, will, motivation and raises the immune system, which has a beneficial effect in the fight against all diseases. Unlike real medicine that really works, placebo effects are short-lived because after a while there is no real effect. The placebo effect gives the best results in the treatment of depression and psychosomatic diseases.

Teams of scientists conducted tests to establish the mode of action of the placebo effect, because it was learned that in some people such treatment produces greater effects, while in others it is much weaker. Studies have shown that the results depend on gender, age, education and many other factors. The tests showed that some subjects achieved excellent results in overcoming

the disease, but it was even achieved that some people feel intoxicated without using alcohol. Results have even been obtained that up to 50% of patients can feel better with placebo therapy. In history, there have been various abuses of this effect (quack doctors, sorcerers, gurus, herbalists), because the ignorance and gullibility of patients were used to achieve material benefit. Today, it is misused for commercial reasons, so non-existent properties are attributed to various drugs in advertising in order to sell better and earn more money. It can be seen from the above that the placebo effect has a very important place in modern medicine and pharmacy and that it is mostly based on the power of belief and autosuggestion (positive energy), but that abuses are always possible (11).

4.3. Bad science through nutrition and antioxidants

Goldacre claims that before nutritionism itself, one should focus on nutritionists who are actually members of a recently invented profession, which must create a commercial space for itself in order to justify its own existence. In fact, in order to achieve this, they have to present the facts about food and the wrong diet to almost every person, in order to turn each and every one of them into a nutrition addict. At the core of their profession is a set of very simple misinterpretations of scientific literature: they make claims about the human race, present data obtained from observations, and finally cite published evidence from scientific research that does not exist.

Various information in the field of nutrition can be found on Internet. A paper was recently published examining the link between modern diets, the obesity epidemic and chronic diseases (such as type 2 diabetes and heart disease). According to them, the results are clear: abandoning the traditional way of eating in favor of modern processed food that has a high content of sugar, refined flour and vegetable oils is one of the most significant factors that contribute to the disease. In the last few decades, total food intake has increased. The reasons for this are different. Among the factors that certainly contributed are the greater availability of food, increased representation of products containing sugar, more aggressive types of advertising of products aimed at children as consumers, etc.

In the 1970s, nutritionists began to advise reducing fat intake. Around the same time, the obesity epidemic began. Of course, it cannot be claimed that the cause-and-effect relationship is firmly established, but the connection between these trends can be explored. Namely, healthy eating guides advised reducing fat intake, which is why sugar intake increased significantly. Intake of processed sugars has increased sharply, reaching up to 67 kg per person per year in some countries. The human body cannot cope with such huge amounts. This poor eating habit leads to numerous disorders, such as insulin resistance, metabolic syndrome, type 2 diabetes, and heart disease. Of all the types of foods that contain sugar, sweetened drinks are considered the most harmful type.

The body does not register calories from solid food and calories from drinks in the same way. Fruit juices are not a better choice. One of the most interesting conclusions presented in the text "11 Graphs That Show Everything That is Wrong with the Modern Diet" is that the wrong attitude of nutritionists led to the epidemic of heart disease. Namely, saturated fat has been wrongly characterized as a cause of heart disease. That is why butter and lard have been replaced by vegetable oils. Vegetable oils, in fact, can contribute to inflammation and various diseases. The fatty acids found in vegetable oils are incorporated into the membranes of our cells, leading to structural changes that can be harmful. Margarine is made by processing vegetable oils so that they remain solid, instead of liquid. The resulting trans fatty acids in margarine actually increase the risk of heart disease (12).

Nutritionists are happy to refer to basic scientific research conducted in the laboratory because it seems that they are actively participating in the process of complex, inscrutable, high-tech academic work. But considerable caution is necessary when extrapolating the reaction of several cells in laboratory conditions to the complex system of a living human being, where everything can function completely opposite to what would be expected on the basis of laboratory work (7).

"Bad Science" also deals with the problem of antioxidants, where he states that in fact an increasing number of nutritionists want to spread the belief that more antioxidants should be taken into the body, where the author does not want to portray antioxidants as something completely unimportant for health, but rather points to the nutritional opinion about antioxidants.

We live in a world full of negative elements and we are surrounded by them from all sides—the air is polluted, we suffer from UV radiation, even our body produces stress hormones and toxic chemical reactions. All these toxins weaken molecules in our body, and chemical reactions create free radicals (molecules without electrons that want to steal electrons from other molecules) or oxidants. Antioxidants are natural substances found in vitamins, minerals and other compounds in food. It is believed that they help prevent diseases by neutralizing free radicals, substances that lead to damage in the body. Free radicals are formed in the body during cellular respiration, but also in a polluted external environment. Free radicals are molecules that damage cells and can contribute to the development of cardiovascular diseases by oxidizing cholesterol. Oxidation of low molecular density lipids (LDL "bad" cholesterol) results in fatty deposits on the walls of the arteries (arteriosclerosis), which slow down or block blood flow to the heart. Free radicals are one of the most well-known factors that lead to the appearance of cancer, they cause inflammatory conditions such as arthritis and are largely responsible for the aging process (13).

The first indicators in favor of antioxidants were really exciting and were more significant than ordinary observational data on nutrition and health. In 1981, Ri-

card Peto, one of the most famous epidemiologists in the world, who is credited with discovering that smoking causes 95% of lung cancer, published a very important paper in the journal *Nature*. He analyzed many studies that seemed to show a positive relationship between higher levels of beta-carotene in the body (an antioxidant from food) and a reduced risk of cancer. The evidence included case-control studies, where people with various types of cancer were compared to people who did not have cancer (but were of the same age, social class, sex, etc.), and it turned out that the people who did not have cancer had higher plasma carotene. This showed that a higher amount of antioxidants might be desirable. After the publication of Pet's work, two large trials of antioxidants were conducted. One was conducted in Finland, where 30,000 participants at high risk of colon cancer were selected, randomly assigned, and given beta-carotene, vitamin E, or both or neither. In addition to more colon cancer among people who received the supposedly protective beta-carotene supplements, compared to those who received a placebo, there were also more deaths from both colon cancer and heart disease in the vitamin group. The results of the second trial were almost even worse. That trial was called the "Carotene and Retinol Efficacy Trial" or CARET, in honor of the large amount of beta-carotene in carrots. Two groups of people with a high risk of lung cancer were studied: in one, smokers, in the other, those who were exposed to asbestos at work. One half was given beta-carotene and vitamin A, and the other half a placebo. It was planned to follow eighteen thousand participants, on average for six years. However, the experiment was terminated early because it was considered unethical to continue. This is because people who received antioxidants pills were 46% more likely to die from colon cancer, and 17% more likely to die from any cause, than patients who received placebo pills (7). This was followed by a review of the quality of the study, which ultimately found that antioxidant supplements were either ineffective or perhaps even directly harmful. This begs the question: Are antioxidants really as they are presented by various experts and do they really help to eliminate diseases and slow down aging?

4.4. Bad science through the use and abuse of medications

Why is medication abuse so widespread? First, they are readily available — practically every home has them. Secondly, many young people are fooled into thinking that they are not doing anything illegal if they take medications that have not been prescribed by a doctor. And thirdly, it seems to them that medicines are less harmful than medications. Some young people think: "If medications can be prescribed to children, then surely they are not harmful." True, when used correctly, medications can improve health and quality of life, and even save life. But if they are abused, they can be as dangerous as medications. For example, abuse of medications with a stimulating effect can cause cardiac arrest or convulsions as in

epilepsy. Some medicines can cause breathing problems and eventually cause death. The medication can also have a harmful effect if it is taken in combination with other medications or alcohol. In early 2008, a popular actor died from a "lethal mix of six types of tranquilizers, sleeping pills and painkillers," the Arizona Republic reported (14).

One of the examples cited in the book "Bad Pharma—How Pharmaceutical Companies Mislead Doctors and Harm Patients" is how GlaxoSmithKline neglected to publish a study showing that the antidepressant paroxetine did not work in children. And this despite the fact that they knew that paroxetine was being prescribed to thousands of children. Not only are children being prescribed a medication that doesn't work, but they are exposed to the side effects of paroxetine. Paroxetine has also been shown to increase the risk of suicide in some children with depression. But paroxetine is only the tip of the iceberg. It is very disturbing that it is not uncommon for studies with negative findings to be buried in this way. Studies funded by the pharmaceutical industry will show positive results much more easily than publicly funded studies. This goes deep: from deciding whether or not to publish something, to manipulating studies by using high doses of the medications being compared as a benchmark or by changing the criteria by which treatment is evaluated, only when access to all the information is gained. Legislators who are supposed to prevent such things are often characterized by either naivety or disrespect for the public interest. That is why, for example, the European Medicines Agency (EMA) has refused to publish the details of a study conducted on the weight loss medication rimonabant for four years.

The European Medical Agency presented a ridiculous argument, that the research data for the medication rimonabant are secret, that is, that it is commercial information, without explaining what requires such secrecy. Eventually, it came to light that people taking rimonabant were at increased risk of a number of serious psychiatric side effects, and the medication was withdrawn. In short, the EMA's refusal to release the information meant that knowledge of the increased risk remained hidden, for inexplicable reasons. The damage done by hiding data and publishing only the most positive studies contributed to the lethargy of researchers, legislators and industry and prevents them from fixing what is supposedly not an obvious problem, such as quack-based health policy. After all, as poor an excuse as it is, we can somehow understand that positive results will always be published rather than negative ones (15).

Modern living conditions have also led to an increase in the negative trend of abuse of prescription medications, or their consumption for non-medical purposes. The problem of medication abuse is still not talked about much, but the fact that it is a significant public health problem all over the world is not hidden (16).

4.5. The downside of pharmacy

Painkillers kill more people than cocaine and heroin combined. The number of deaths caused by overdoses

of prescription painkillers has tripled over the past ten years, according to a report by the Center for Disease Control and Prevention (CDC) in the US. New findings show that more than 40 people die every day due to overdose. *“Overdoses involving the use of painkillers are now reaching epidemic proportions, killing more Americans than cocaine and heroin combined”*, said Dr. Thomas Frieden, director of the CDC. In 2008, it was estimated that painkillers killed more than 15,000 people that year, but this statistic, which includes a decade's worth of data, reflects the lethality of the medication industry.

The “legal” trade in medications and opiates appears to be more deadly than the “illegal” as the 415% increase in fatal overdoses of painkillers is four times the total number of deaths caused by cocaine or heroin overdoses. This means that, as medications are increasingly accepted in society as a way of life (or a way of legal medication use), the number of analgesics increases without increasing the actual reasons for their use. It's amazing that pharmaceutical companies and governments are fully aware of this fact, but still supply patients with painkillers. Of course, it is difficult to determine which of the patients really need an analgesic, but it is not difficult to understand that something is wrong with the millions of extra painkillers produced every year without any basis in real conditions. Research published in the international journal Medication Policy found that the abuse of such medications could be a way to avoid hard and illegal medications. The researchers focused on teenagers who abused both injectable and pill medications. 85% reported using opioid painkillers before trying heroin. It is reported that, on average, the use and abuse of painkillers begins about two years before the use of heroin. It seems that the pharmaceutical industry is once again out of control, and socially accepted “legal” medications are more dangerous than the illegal alternative. And, it seems, there are users of legal medications who die more often than users of cocaine or heroin (17).

There is a famous case for the “medication” OxyContin. It is a narcotic analgesic that has fueled the problem of medication addiction over the past few years. This “medication” is highly addictive and leads people to use heroin when they can no longer afford the overpriced OxyContin. This problem became even more pronounced after the manufacturer decided to change the composition of the medication to allegedly discourage addiction in users of the “medication.” OxyContin was originally designed as a medication that was released into the body's system slowly over several hours, meaning that each pill contained a large amount of oxycodone. Medication users soon discovered that by crushing and snorting or dissolving the pill in water and injecting it directly, they could bypass the slow-release mechanism. In 2010, Purdue Pharma changed the formula to make the tablet harder and to dissolve more slowly. A significant percentage of former OxyContin addicts have since turned to more powerful medications such as heroin and other stronger opioids. The creator of this miracle medi-

cation, responsible for countless overdose deaths, is now worried about the patent expiring and has decided to test it on six-year-old children. The company in question is Purdue Pharma LP whose motives are not helping children and overcoming deadly disease or reducing pain. By starting to test on children, Big Pharma can extend the validity of the patent by six months. This is possible through an FDA program that encourages medication companies to test their poisons on the youngest members of our community. A company is willing to put children in a lab in exchange for just six months? But for Purdue, like many similar companies, that's not a concern. In 2011, Purdue earned 2.8 billion US dollars from the sale of OxyContin, while a six-month period carries potential earnings of almost another billion and a half dollars (18).

4.6. The “case” of Thalidomide

An extremely poignant example of a new therapy that caused more harm than good is the medication thalidomide. This sleep medication was introduced into medical practice in the late 1950s as a supposedly safer alternative to the barbiturates that were regularly prescribed at the time (19).

Newly discovered documents reveal that some 50 years ago, pregnant women around the world were used like laboratory guinea pigs to test the effects of the anti-morning sickness medication, thalidomide, in a series of clinical trials that resulted in the birth of children with severe birth defects. Thalidomide is a sedative medication introduced in the late 1950s, which was used as a sleeping pill, but was soon found to be effective against morning sickness. It was sold from 1957 to 1961, when it was withdrawn after it was found to be teratogenic, meaning it caused birth defects.

The medication was developed by the German pharmaceutical company Grünenthal in Stolberg. Heinrich Mückter, a former member of the Nazi party and an army doctor, had headed Grünenthal's research department since the company was founded and was responsible for the creation of thalidomide. On October 1, 1957, thalidomide was launched, and it was claimed to be an effective sedative and pain reliever, and was hailed as a “miracle cure” for insomnia, coughs, colds, and headaches. It has also been found to be an effective antiemetic that has an inhibitory effect on morning sickness, and thousands of pregnant women have taken the medicine to relieve symptoms. At the time of the medication's development, scientists did not believe that the medication taken by pregnant women could pass through the placenta and harm the development of the fetus. It is estimated that as a result of taking thalidomide during pregnancy, in 46 countries of the world, between 10,000 and 20,000 children were born with deformities. Some countries such as Austria, Switzerland and the US have not approved the medication's use, but although thalidomide was never approved for sale in the US, millions of tablets were distributed to doctors during the clinical trial program. Finally, around 1961, the medication was

withdrawn from free sale, but today it is available on prescription, and is administered under strict supervision. In 1960, Australian trials were the first of their kind in the world, where pregnant women were given Distaval, a thalidomide medication, according to *The Age*, an Australian daily newspaper. According to *Natural News*, documents obtained from thalidomide distributor Distillers Biochemicals Ltd indicate that Australian women were given Distaval before tests were done on domestic laboratory animals to determine the effect of thalidomide on fetuses. A letter written by Distillers' management dated September 1962 confirms that: "*no tests were carried out regarding the effect on pregnancy before the medication was placed on the market*". According to the documents, studies on the medication were conducted by obstetricians from "Sydney's Crown Street Women's" and "Royal Prince Alfred" hospitals as early as May 1960. Hospitals in Melbourne and Adelaide were also conducting trials, but it remains unclear whether pregnant women were given Distaval there. A company letter dated July 1960 records the receipt of "an urgent telegram from the Royal Melbourne Hospital, requesting additional deliveries of Distaval." It also mentions "further success (of medication testing) from the Royal Prince Alfred Hospital in Sydney." The records also show that the dangerous medication was released to the general public around August 1960, and was touted in advertisements as a medication of "exceptional safety."

In 1961, babies with deformities began to be born to women taking thalidomide. The records also show that several of the company's employees in the UK, Australia and New Zealand were among those whose children were born with defects from the medication. Sydney obstetrician William McBride, who participated in the trials, contacted distributors twice between April and July 1961, warning that women taking the medication had given birth to deformed or stillborn. "*I have had four deaths in childbirth in the last eight weeks,*" he writes. "*One baby was born with six toes, one with clubbed toes and one with deformed toes*" (20).

As reported by The Guardian, in 1972, the Sunday Times published a front page with the slogan "Our thalidomide children, a cause for national shame", as part of a long campaign for additional benefits. In the end, during the 70s, a total of 28 million was paid out. pounds compensation from Diageo. In 2005, a Kenyan boy without arms and legs was granted a visa to travel to the UK to receive medical treatment following a campaign by the charity Thalidomide UK. It is not known what caused the child's disability, but the association claims that the medication is still used to treat leprosy and AIDS in developing countries. Sadly, the 14-month-old boy dies of a yeast infection the following year. Meanwhile, Diageo is agreeing to more than double compensation payments to thalidomide victims from £2.8m a year to around £6.5m. All this is not enough, and the medication is still used, new possibilities are being explored: In 2007, a study showed that thalidomide can

significantly improve the survival chances of patients with bone marrow cancer. Researchers say adding thalidomide to standard treatment extends life in elderly multiple myeloma patients by an average of 18 months, and in 2008 the medication was approved for treatment of multiple myeloma by the European Medicines Agency (20). Dr. Judah Falkman, who first used thalidomide in the treatment of multiple myeloma, believes that thalidomide prevents tumor-induced angiogenesis. It does not lead to a cure but to a better control of the disease. Unlike many anticancer medications, thalidomide has multiple effects on cancer. As it affects angiogenesis, it may directly affect cancer cells and activate the immune system against the disease.

But, in other parts of the world, the disaster happened again. Between 1970 and 1980, new generations of babies were born with the consequences of thalidomide use - this time in Brazil, where leprosy is a problem for many residents and where thalidomide use is widespread. Some women did not understand the warning about the dangers of using the medication because it was written in English. <http://www.cigota.rs/vest/thalidomide-medicationa-sansa>

In 2009, scientists from the University of Aberdeen claimed to have solved a "50-year puzzle" after discovering that thalidomide causes limb loss. They discovered that an ingredient in the medication prevents the growth of new blood vessels in developing embryos, preventing limb growth. The Government has pledged to pay £20m to the thalidomide fund over three years, following another campaign by the Sunday Times.

In a speech at the memorial gathering in memory of thalidomide victims in Stolberg, 31 August 2012. Ph.D.Sc. Harald F. Stock, Ph.D. sc., the chief executive officer of the Grünenthal Group, issued a public apology from the company, but stated that the company acted in accordance with the scientific knowledge and industry standards at the time. Victims of thalidomide, the medication that caused thousands of children in the 1950s and 1960s to be born with severe deformities, announced that the apology from its manufacturer, the German company Gruenenthal, was "dishonest, belated and insufficient". The company that patented thalidomide, for the first time in 50 years, she apologized to the victims of the medication, which in 1957 she claimed was "as safe as a sugar cube" (21).

Pharmacologist Frances Kelsey refused to sign the official approval of the US Food and Medication Administration (FDA) and saved thousands of children's lives by doubting the safety of the medication thalidomide. For this reason, in 1962, John Kennedy presented her with a special presidential award. Thalidomide is actually a synthetic chemical that exists in two different forms that relate to each other like mirror images (like left and right hands). But while one form acts as an antiemetic (against nausea and vomiting), the other has a teratogenic effect on the human fetus. Scientists have been trying unsuccessfully for years to discover the molecular mechanism

by which thalidomide disrupts the normal development of unborn children. It was only in 2010 that Japanese scientists were able to do this, who attached the thalidomide molecule with a special magnetic marker, inserted it into a cell culture and detected the protein cereblon, which is a specific “victim” of the interaction with this medication.

While Japanese scientists believe that they have solved the mystery of thalidomide's toxic effects, other scientists believe that this is just the beginning of interesting discoveries. New Zealand professor Ching believes that the Japanese study is only the twentieth scientific model that interprets the teratogenicity of thalidomide.

4.7. The role of the pharmaceutical industry and the importance of cholesterol treatment with statins

Of all the victims of the bad and biased science of the past decades, cholesterol and saturated fat have paid the most, and they still do. Because someone had to be a victim in order to satisfy the needs of the public and the needs of the multibillion-dollar pharmaceutical industry. Cholesterol is derived from the ancient Greek word *cholestereos*, where *chole* means “bile” and *stereos* means “solid”. Today, millions people on Earth use statins, which are medications to lower blood cholesterol (17, 22). Although there is no credible study confirming the effect of statins on reducing heart disease, doctors prescribe them as if they were lemon water. In the 1980s, there was a revolution and since then statins have become the main group of medications, earning up to 25 billion dollars a year. Statin use on a monthly basis can be as much as \$200, and on the list of different brands of statins next to many there is an indication that there is no or unknown association with the reduction of heart disease. Statins are HMG CoA-reductase inhibitors, and their role can be clearly seen. It shows the whole process of how cholesterol is produced, and that process is branched and can be represented as a tree. The pharmaceutical industry has judged it wise to get rid of the cholesterol (dried leaf) by interrupting the process between HMG CoA-reductase and mevalonate (mevalonic acid) by cutting down the tree, even though this process is many steps behind the final process of obtaining cholesterol. Of course, statins were created as HMG CoA-reductase inhibitors because experts did not know how to stop the process of obtaining cholesterol from squalene, which would have the least negative effects. So, you should not think at all that the function of statins can pass without negative consequences, because it leads to the interruption of the synthesis of coenzyme Q10. In addition to blood cholesterol-lowering medications, cholesterol is blocked, coenzyme Q10, dolichol and farnesol and protein-geranyl geranylation and dimethylallyl etc. are also blocked. And coenzyme Q10 is essential for heart health and for the proper functioning of the brain and muscles. This is why the major negative effects of statins are muscle pain, weakness, memory loss. Nowadays, about 85% of studies are funded by medication companies, and then there is a 24% greater chance that

those studies will show that the medications are effective, and an 87% greater chance that no negative effects will be reported. The studies present valid data, but the problem is that they are done and designed so that they do not present all the negative effects, but there is a lot of manipulation and bias. The problem is that the studies and the hypotheses in the studies do not ask the right questions.

Two of the biggest fallacies of science when it comes to heart health, nutrition, and cholesterol are: a) Eating foods rich in cholesterol increases blood cholesterol; b) High blood cholesterol levels cause heart disease. Neither claim has anything to do with reality, and those claims are the result of bad science riddled with bias. Contrary to popular belief, cholesterol is not harmful and does not cause heart disease. A diet high in cholesterol does not increase blood cholesterol levels. Cholesterol is an essential organic product that makes up the cell membranes of all mammals on Earth. Cholesterol has very important roles in our body, the most important of which are enabling the body to produce sex hormones (estrogen in women and testosterone in men) and vitamin D, production of bile acid without which digestion would not be possible, the aforementioned property of forming cell membranes, etc. A reduced level of cholesterol in the blood is associated with various diseases, so it is very important to maintain a diet that contains cholesterol. A study done in Japan on 47,000 people concluded: “The highest death rate was found in those who had the lowest cholesterol of 160 mg/dl (4.14 mmol/L), and the lowest death rate in those who had higher cholesterol values of 200 to 259 mg/dl (5.17 to 7 mmol/L)”.

4.8. The role of expert systems in diagnostics and therapy

The knowledge base is a collective archive of the knowledge and memory of several specialists and experts. It contains objective laws or rules for solving problems and unspecified heuristic facts that the expert has come to in the course of long-term practice and which he knows will significantly help him in solving similar tasks. Production rules are used in most expert systems (23-25). Each of the rules has a premise and an action which are used for a wide range of medical problems: diagnosis and treatment of infections, neoplasms, glaucoma, cancer staging and prognosis, interpretation of signals and test results, prediction of drug interactions, etc. Biochemistry is the science of the chemical composition of living things and of chemical changes in living things. Biochemical - laboratory diagnostics occupy a prominent place in medicine. Laboratory diagnostics is a very important activity in health care. Laboratory diagnostics in primary health care is software that provides short and useful information as an aid in understanding, choosing and interpreting laboratory tests. Today's knowledge in the field of laboratory diagnostics enables reliable diagnostic verification of the physiological and pathological condition of the subject and monitoring of the patient's therapy. Today's knowledge in the field of laboratory di-

agnostics enables reliable diagnostic verification of the physiological and pathological condition of the subject and monitoring of the patient's therapy. There is a high level of knowledge about the metabolism of individual constituents in the body, and their monitoring is technically very accessible. Therefore, with modern laboratory diagnostics, using different systems and reactions, the doctor can obtain very useful data about different flows of normal, pathological or metabolism under the influence of drugs. Today, it seems that the patient is more in touch with instrumental techniques. The use of different systems and reactions with modern laboratory diagnostics, the physicians can obtain very useful data about different flows of normal, pathological or metabolism under the influence of drugs. Laboratory diagnostics has not lagged behind in the application of informatics, because every device in the laboratory is monitored by a computer and IT software that is exclusively related to a certain diagnostic search or a certain procedure.

Clinical and polyclinical laboratories use automatic interpretation of test results by a computer system and one of the most successful areas in which expert systems are applied. The first expert systems in medicine appeared during the 1960s, only to begin to record their true expansion during the 1970s (23). Some of the older expert systems exploited in medicine were: MYCIN, CASNET, Internist, Internist II, PIP, Digitalis Therapy Advisor, EXPERT, Ledi-2, UMDES, MODIS-2, JOSEPH, ONCO-HELP, PHARM-2, QUAWDS, etc. Of the newer expert systems, the following are in use: GermAlert, Help, Thoraks, Derma Dex, Pepid, ADE, CCIS, FACTS, Geninfer, Larsen, Mammonet, Acorn, Cadiag 2, TDW, ICONS, Diagnosis Pro, CMD, MSO, etc. Modern expert systems are based on www technology, which enables their wide use, or availability to a large number of users, as well as the possibility of integration into the global (world) network of ES - a.

DENDRAL is one of the first ES developed at Stanford University, which is used in the field of chemistry to determine the structural formula of organic compounds (23). The role is to assist the researcher and systematically consider all possibilities in accordance with the imposed restrictions that narrow and direct the search for candidates. GermAlert and GermWatcher are expert systems that scan microbiological cultures from the hospital's laboratory systems, and based on the obtained results, warn of the possibility of infection and the need for adequate treatment. A more general example is the PEIRS system, which on average interprets about 80-100 reports per day with a diagnostic accuracy of about 95%. Interprets thyroid function tests, arterial blood gas concentration, catecholamine content in urine and plasma, hCG (human chorionic gonadotropin) and alpha - FP (alpha fetoprotein), glucose tolerance tests, cortisol, gastrin, cholinesterase, etc. HEPAXPERT - III is a rules-based system that stores and interprets the results of serological tests for infection with hepatitis viruses A, B, C and D. Interpretation of acid-base disorders is an ex-

pert system that, based on data on the content of respiratory gases in arterial blood and serum electrolytes, talks about acid-base balance disorders. Liporap includes a system that, based on data such as: total cholesterol, TG, HDL, LDL, VLDL, phospholipids and similar - classifies lipoprotein disorders according to Frederickson, or as much rarer diseases (Tangier, LP - X ...), and provides additional information, such as chronic diseases and medications, that could explain the patient's condition. Pro.M.D. is the shell of an expert system introduced in 1986 and refers to lipoprotein disorders. More than fifteen Pro.M.D. are in use in clinical practice. systems that include EBV serology, thyroid hormones (Thyrolab), interpretation of alkaline phosphatase, interpretation of protein in urine, lipoproteins in serum, etc. SahmAlert serves to evaluate the effectiveness of antibiotics. Using criteria developed by local epidemiologists, the system scans data on microbiological cultures, thereby identifying those containing microorganisms with unusual resistance to antibiotics. PharmADE monitors patient care by screening medications that could interact with medications already given to the patient. Becton Dickinson, an international company, has developed several systems for routine use. One of them is the so-called MIC Sceptor - an interpreter that detects the minimum dose of antibiotics needed to destroy microbes. The system is primarily designed for use in hospital microbiology laboratories. MYCIN is an ES developed at Stanford University in the early 1970s, and is intended to determine antibiotic therapy for the treatment of more severe bacterial infections (23), such as bacteremia and meningitis. Suggesting therapy is based on diagnosis, in this case determining the bacteria causing the infection. The causative agent of the infection is determined based on the results of laboratory tests of the patient's blood samples. This data can be, for example, related to blood samples taken and analyzed, bacteria identified from the samples, antibiotics that work against these bacteria, data related to previous infections, operations, etc. CADUCEUS is a medical expert system created with the intention of improving the MYCIN system, in terms of covering a wider field of medicine, so that it could ultimately diagnose more than a thousand different diseases. DoseChecker is an expert system that automatically evaluates prescriptions for a certain group of drugs in terms of their inadequate dosage and the possible need for dose adjustment, based on specific information about the patient. This would primarily include patients with kidney damage, so the patient's body weight, serum creatinine, and creatinine clearance are used as parameters. The listed expert systems are some of the most commonly used in biochemical diagnostics. There are a lot of other expert systems in the praxis. But, some of them still in use but improving of its systems are not appropriate. The author of this text actively participated in discussions at the notable examples referred to the Expert Systems, which EASA President Academician Maizner spoke about it in his presentation, emphasizing their application in

clinical work, to which I asked his opinion: "Why expert systems, which were intensively used in the seventies and eighties, in last century and they literally stopped in development and some were almost forgotten"? Prof. Mainzer thinks that maybe, in his opinion, there are two key reasons - the dysplosion of the use of ICT in the last 20 years, and Big Data, which "occupied" almost all fields of science and profession. Mainzer's answer was in agreement with my opinion, which are the key reasons for less and less training, upgrading and the use of already developed expert systems, and only those that appeared in use in the last two decades. But, my opinion that reasons are, also, economical nature, because some of them not provide enough money and no motives to spent new resources for create new versions of expert systems.

4.9. COVID-19 Vaccines-Medical and Ethical Aspects

As Editor-in-Chief of a few medical journals I received a lot of papers with subjects about corona disease, but definitely, locally and globally, corona pandemic and its consequences (health protection and treatments, social, economical, educational aspects, etc.), in almost every country dropped to the great crises. It is unclear and probably we will wait for explanation, maybe 5-20 years in the future, what will be final corona consequences. Maybe it will be similar to cases described in the text before. Nobody exactly now what kind of diseases will be provoked by the corona virus in our tissues and organs in the body or what will be influenced on diseases which everybody of us already have currently in our body. Probably, morbidity and mortality rates and official statistical pictures and statistical report lists will be dramatically changed in every country in the world.

There is not so much realized and published EBM studies in the scientific literature worldwide. distributed until now. One of the relatively systematic cross-sectional study have done by Abou Ghayda R. et al, published in Medical Virology Journal (I am one of co-authors), but presented statistical meta-analysis is already old, but collected data are not enough exact (29). Until now probably over 70 variants of corona viruses existed in the world and, also, almost every country have some kind of vaccine. Reports about symptoms of corona infections were changing day-by-day during last the 3 years - from cytokin storm in the first half of 2020, later vascular problems, gastrointestinal problems, "brain fog" and mental problems and finally mixed symptoms, and experts telling about "long covid". Because of that we announced hypothetic theory that probably corona virus will have behavior something like B and C virus of hmpatitis, and which will move from one to another place/part in our body and wait when body lose its immunity and - to be attacked once again. A lot of common people, but especially medical professionals, as the first risk line, a few times were infected, besides vaccinated. Dilemmas about efficacy of vaccination are day-by-day more and more actual (30). But decision maker(s) - WHO and their experts until now kept words of explanations what are basic and capital attitudes. Although the benefits of vac-

cination for a society are obvious, a non-negligible percentage of people are refusing this protection against COVID-19.

In the United States, state policies mandate certain immunizations, including school entry requirements, which cover significant numbers of children (31, 32). The first school vaccination requirements were enacted in the 1850s to prevent smallpox (33). Federal and state efforts to eradicate measles in the 1960s and 1970s motivated many modern mandates policies (32) including discussions about vaccine development, and study design, population, and trial location. Some states have specific informed consent laws (31). Certain lawmakers and other patient rights advocates believe that requiring specific consent is ethical and appropriate, so that parents are better informed about vaccines, and have adequate time to ask questions if needed (32). Opponents fear that a regulated written consent procedure may add unnecessary fear or concern to the vaccination process (31). To be licensed, vaccines undergo many years of research, and must pass rigorous safety and efficacy standards (34). The vaccine development and research process includes diverse experts of many scientific and social disciplines, including public health, epidemiology, immunology, and statistics, and from pharmaceutical companies. These stakeholders may have conflicting priorities and motives, which contributes to various ethical discussions (30). Examples of disparities in the United States and globally signal the need for continued efforts to ensure equal opportunities for people to benefit from vaccination (31-34). Ethical discussions also surround the research and testing of vaccines, including discussions about vaccine development, and study design, population, and trial location (30). Ethical debates also surround vaccine implementation and delivery, such as those concerning informed consent (33, 34). Some states have specific informed consent laws (31). Many vaccine-related ethical debates center on the evidence that access to vaccination depends in some way on socioeconomic and racial ethnic minority status. Implicit in these discussions is whether all lives are equal, and equally deserving of opportunities to be protected by vaccination (30).

Certain lawmakers and other patient rights, led by made headlines over the last 10 years in US, advocates believe that requiring specific consent is ethical and appropriate, so that parents are better informed about vaccines, and have adequate time to ask questions if needed (34). To properly test a vaccine's effectiveness, a clinical trial including a control group that does not get the test vaccine is usually necessary (31). However, failing to provide any adequate preventive option can be a difficult decision when the vaccine can potentially prevent a serious, untreatable, or fatal infection. It is important to understand a vaccine's safety and efficacy in various populations, but testing a vaccine in vulnerable populations, such as children, also raises ethical concerns (32).

Farouq Ahmad Faleh Alazzam et al. discussed about

COVID-19 vaccination dilemmas and its medical, ethical and legal aspects (30). A lot of dilemmas opened for discussion during last three years and consequences which are day-by-day more and more attractive for discussion. In February 2020, WHO, UN and Verified in US made partnership and launched an initiative to cut through the noise and make sure accurate, fact-based information about COVID-19 was reaching people early and often. vaccines undergo many years of research, and must pass rigorous safety and efficacy standards (36).

The COVID-19 pandemic created an "infodemic", because misinformation was causing confusion and increasing the spread of the virus, so we developed content to increase digital literacy and help people recognise misinformation before they shared it, as stated Jen Thomas (Creative Produced in Verified) in his text *"Fighting misinformation where it happens: striking first and often"* (36). "Misinformation can feel like an insurmountable problem, something that only tech giants and social media platforms can solve. Their initial campaigns focused on how the virus spreads and what they can do to prevent it, such as social distancing and wearing a mask. When vaccines started to become available it was created simple explainers in response to some of their audience's most pressing concerns, including how the vaccines work", why they know they are effective, and why vaccination is important. *"But actually, we can each take steps to help stop it spreading, and these actions will have a direct impact on our own communities, friends and family. Misinformation was causing confusion and increasing the spread of the virus, so we developed content to increase digital literacy and help people recognise misinformation before they shared it"* (36). By him, *"the accidental spread of inaccurate information, makes it harder for people to find accurate information online. It can also cause fear, panic and hate, making difficult situations even more challenging. The COVID-19 pandemic created an "infodemic", an overwhelming increase of misinformation that has made public health measures more difficult"* (36).

In its first year of work, Verified reached over a billion people worldwide with more than 5000 pieces of content in 60 languages helping to limit the spread and harm of COVID-19 (36). It has become a model for how the UN and our global partners can respond to emerging global challenges quickly and effectively. One another text about fighting from misinformation was: *"It's the news that everyone has been hoping for. Pfizer announced that its late-stage vaccine for COVID-19 not only works more than 90% of the time in humans but also has no harmful side effects. The information sent stock markets soaring, thrilled scientists, and gave hope to hundreds of millions of people that there could be an end in sight to the coronavirus pandemic. But nothing about COVID comes without complications"* (36).

Following the consideration of COVID-19 as pandemic by the WHO, developing new vaccinations against COVID-19 was the dream of humanity (29). Leading

companies competed to achieve this task. Several vaccinations have been developed relatively quickly. The main findings showed that a dilemma exists in literature regarding the ethics in general in keeping the rights of individuals to retain their rights to receive the vaccine and considering receiving the vaccination as compulsory (30). As the disease has become pandemic with high mortality rates, keeping the safety of the community has received the priority on individual rights, and many countries considered compulsory vaccinations. It has been determined that the most effective preventative measure that is currently available is the vaccination against COVID-19 (25). This has led to fewer deaths and hospitalizations, which has made it possible to combat the pandemic by combining it with non-pharmacological interventions. Additionally, this has resulted in the pandemic being able to be contained. Even though most adverse reactions to vaccination present with mild symptoms and serious effects are extremely rare, individuals who have experienced them have the right to take legal action against the healthcare workers (HCWs) who administered the vaccination. This is the case even though most adverse reactions to vaccination present with mild symptoms. In order to find information on vaccine injury compensation programs, new laws or policies to protect HCWs administering vaccinations that were introduced during the COVID-19 pandemic, and policies on mandatory vaccinations, it has been conducted a search for the three most populous countries on each continent (30, 32). Because of the gravity of the situation, every country on the planet is contributing to the care of COVID-19 patients (29). The network of health care workers who were a part of the emergency management system served as the foundation on which the response to the COVID-19 outbreak was built. As a consequence of this, it is of the utmost importance to preserve their physical and mental health to the fullest extent possible, in addition to making certain that they are protected legally and by regulatory authorities (30). Most countries continue to have a low vaccination coverage, even though the COVID-19 vaccination is one of the safest and most effective measures to prevent infection, hospitalization, and death. Indeed, the dissemination of misleading information and breakdowns in communication helped contribute to the growth of a distrust not only of vaccinations but also of the medical professionals who give them the adverse effects of COVID-19 and legal framework (30).

The COVID-19 vaccination can cause some unwelcome side effects in some people, just like any other medical treatment. This is true for both children and adults. Even though most of these reactions exhibit mild symptomatology and severe effects are extremely uncommon, it is important to keep in mind that they can give rise to legal action against the healthcare operator who administered them (29). Both national and international institutions have made efforts to increase vaccination coverage and confidence in the COVID-19 vaccination by promoting

vaccine injury compensation programs, which are also known as VICPs (31). At the same time, specific laws have been enacted to protect physicians from the legal responsibility for injuries caused by the administration of vaccines. These laws have been enacted in several countries (like France and Italy) which have made use of these tools, establishing a specific compensation system for COVID-19 vaccine-related injuries and the latter nation establishing a criminal shield (30). World Health Organization (WHO) and the United Nations worked to increase awareness of the COVAX No-Fault Compensation Program on an international scale as an example of a VICP, and its objective was to provide high-risk and vulnerable populations with vaccines that are both safe and effective against COVID-19 (30).

Lee ACK and Morling JR (35) professionally explained dilemma about COVID-19 vaccine, and stated: *"There is hope - the wonders of modern vaccine science have seen the rapid development of more than 68 vaccines worldwide, and around ten have received emergency authorisation and use thus far. This has opened a new front in the struggle to control the pandemic, offering the potential to achieve population immunity through vaccination. Vaccination is far safer than the more hazardous route of achieving population immunity through natural infection that carries a high risk of COVID-19 mortality and morbidity. Early experience with the vaccine is positive when compared directly with the effects of COVID-19 infection. For example, the roll-out of the Pfizer-BioNTech vaccine in the US observed only 21 cases of anaphylaxis after administration of nearly two million first doses of the vaccine, with no fatalities reported. This compares favourably against the COVID-19 infection-to-fatality ratio, estimated at around 1.15% in high-income countries. There is also a significant morbidity risk with COVID-19, including the risk of 'long COVID', that is as yet poorly understood. UK estimates are that around one in five persons infected with COVID-19 exhibit symptoms for a period of 5 weeks or longer, and one in ten respondents have symptoms for over 12 weeks a population health perspective, there can be no rational reason for pursuing a population immunity strategy through natural infection now"* (35). They also commented dilemma about doses of vaccines which will be in use against with a rapidly spreading third wave of infections in December, partly driven by the emergence of a new variant (SARS-CoV-2 B.1.1.7), and propose to generate considerable uproar among primary care physicians involved in the delivery of vaccinations for a variety of reasons, including the turmoil and workload associated with having to consent patients and rebook hundreds of thousands of appointments (35). By Lee and Morling the first dose vs two-dose prioritisation saga is also an ethical dilemma for clinicians and they thought that clinicians typically strive to do their very best for individual patients and see it as their moral duty to do so, and giving two doses as per the vaccine authorisation and trial protocol could be seen as the 'right' thing to do (35). Another vaccine dilemma that has emerged is the

decision as to who gets immunised first. The US and UK have both focused initially on the older age groups owing to their risk of mortality. One modelling study supports this approach and found that vaccine prioritisation for the elderly saves the most lives, although vaccinating the elderly may reduce the number of deaths and hospital admissions, this age group accounts for only a small proportion of infections (35). Public health institutions and family medicine teams need to provide a degree of protection to more vulnerable unvaccinated individuals. Finally, the arrival of COVID-19 vaccines has sparked a vaccine race between countries to immunise their populations - race favours high-income countries, and there are real concerns that vaccine nationalism with possibility to create losers and widen global inequalities (it was a case with Bosnia and Herzegovina); ratio of mass vaccination in high-income countries and less vaccination in lower income (poor) countries where the virus was (with higher rate of endemic).

A lot of questions are opened - moral, ethical, legal dimension, etc. A lot of experiences described by professionals out of medicine field, were included in solving of un-equalities between rich and pure countries and their citizens, but main decision makers need to decide. BUT WHO THEY ARE?

Lee and Morling, as Public health experts have opinion: *"should vaccine access not be determined by need rather than national wealth and influence? This is perhaps why the COVAX initiative is vital to ensuring equity of vaccine access.12; mIn an interconnected globalised world, all our fates are intertwined. Global solidarity is needed to protect our national health, wealth and human rights. In essence, we are not safe until we are all safe"*.

When I studied at LSHTM in London in 1981/1982 on blackboard in the GoldSmith theatre the first sentences were written: "If I have information, I have power". Now is different: "If I have misinformation, I have power". But the best is, If I have BOTH and money I have power. *Sapienti sat*.

5. DISCUSSION

One of the most important questions in science is how to know if a certain method of treatment is effective. Using the example of homeopathy, or the method of alternative medicine, Goldacre himself explained that the role of medicine, and people's beliefs about the body itself and treatment, as well as evidence about the good and bad sides of treatment methods are important (7). Goldacre claims that before nutritionism itself, one should focus on nutritionists who are actually members of a recently invented profession, which must create a commercial space for itself in order to justify its own existence (7). In fact, in order to achieve this, they have to present the facts about food and the wrong diet to almost every person, in order to turn each and every one of them into a nutrition addict. Nutritionists are happy to refer to basic scientific research conducted in the laboratory because it seems that they are actively participating in

the process of complex, inscrutable, high-tech academic work. One of the best examples of bad science is the "case" of thalidomide, a "medication" that caused more harm than good. In the end, we can say that in contrast to pseudoscience, that is, bad science, in science there is a standardized procedure by which to arrive at recognized scientific truths and by which a scientific view of the world is formed.

In its first year of Corona pandemic, Verified reached over a billion people worldwide with more than 5000 pieces of content in 60 languages helping to limit the spread and harm of COVID-19. It has become a model for how the UN and our global partners can respond to emerging global challenges quickly and effectively (36). There are many truly serious problems with how big pharma distorts the evidence for prescription medications. Pharmaceutical companies, like all other commercial enterprises, are profit-driven in their practice. Therefore, they do not produce unprofitable medications, although they could save millions of lives in the Third World countries, but expensive and chargeable ones, even if they are useless. About fifty years ago, the pharmaceutical industry was realistically sized and could treat diseases and make money at the same time. In the meantime, on the fertile soil of medicine and healthcare where no one set limits, it grew to such proportions that it "consumed" really sick people. At the current level of expansion, further growth in profits requires the maintenance of existing and the creation of new diseases. Treating people has thus become, in fact, "mission impossible" for the pharmaceutical industry. People are noticing the weaknesses of modern medicine and the awareness of its bias is growing.

In the West, its critics are increasingly numerous, both among scientists and doctors, as well as among laymen-journalists and the public. Associations of consumers of health services are being established that question medical practice, science and education, demand changes and devise alternative ways of treatment and alternative health systems. Medicines are actually advertised on television, followed by shots of happy people drinking them, and at the end of the comically accelerated explanation that the side effects are heart attacks and the like. An analysis of medication advertisements on the US market showed that most of them mislead potential customers by offering them false or half-truthful information.

The worst forms of intellectual dishonesty in scientific research and publishing are scams that are manifested as follows (37-40): a) falsification (falsification of results); b) falsification-if the data does not reflect the actual amounts found in the study; c) fabricating (inventing results); d) fabrication - when the data are completely fabricated; e) plagiarism (plagiarism of the results); f) plagiarism - stealing other people's ideas, concepts, texts; and g) misrepresentation as well as their own, and some other forms described in a lot of my papers in indexed journals (37-40).

Study published in the journal *Psychological Science* has shown that over time, customers ignore medication warnings and pay attention only to the highlighted benefits of the product. While it is true that the fines against pharmaceutical companies are very high, it is nothing compared to their profits. Such corrupt companies pay fines without major problems and continue with their activities. Regardless of all the arguments of one side or the other, there will still be both paths along which human knowledge moves; in addition to the scientific, its adherents will also have an esoteric understanding of the world. However, this will not stop science from advancing; it will continue to bear fruit that will be used and abused.

Scientific institutions and universities should, in accordance with the principles of Good Scientific Practice (GSP) and Good Laboratory Practices (GLP) have a center for monitoring, security, promotion and development of quality research. In this way, internal quality control ensures that a research institution such as a university, assume responsibility for creating an environment that promotes standards of excellence, intellectual honesty and legality. Scientific way of thinking and the application of scientific method require honesty, criticism, confidence, creativity, openness, and acceptance of these principles are desirable prerequisite for successful engagement in science for students and young researchers. Additionally, the principles of ethical codes that govern the broader ethical issues in all aspects of scientific research, the rule in research laboratories (GLP) defining criteria for the monitoring and the basic principles of quality assurance in scientific research (39, 40). Authors and publishers have ethical obligations. In publishing the research results, the investigators are obliged to preserve the accuracy of the results (1, 2, 37, 38).

Scholars have been entrusted to regulate their own activity and that is more than in any other area of human activity. Adherence to ethical codes of good scientific practice (GSP) in institutions of science (1, 2) ensures the quality of teaching. After the outbreak of a series of scandals-deception in science in European countries these rules are binding in all scientific institutions.

Development of science, increasing the number of scientists and scientific publications has increased and the number of occurrences of scientific dishonesty. In the early eighties of the 20th century, the question of scientific fraud becomes public in the western world, especially with more and more cases of scientific fraud in the United States. But, today it is one of most important scientific problems globally. Last 6 years Academy of Medical Sciences of Bosnia and Herzegovina organized Special Topic Conferences with topics: Scientometry, Citation, Plagiarism and Predatory in Science Publishing, with participants who are editors from Balcan countries and, also, worldwide. Presented papers are deposited on [www:amnbih.org](http://www.amnbih.org). Academy, also, published "Sarajevo Declaration on Integrity and Visibility of Scholarly Publications" and "Guidelines for Editing Biomedical Journals"

as manual instructions (41-45).

Also, reputable international journal editors from different scientific fields established association COPE (Committee on Publication Ethics), which, in addition to promoting the scientific principles of fairness, provides guidelines and suggestions for the editors of scientific journals on the procedure in cases of suspected dishonorable actions in published studies and papers submitted for publication. According to the recommendation of the International Commission for the professional self-regulation of science the rules of good scientific practice are related to the basic principles of scientific research.

During the last 30 years, there has been a rapid decline in the level of science and education in Bosnia and Herzegovina, and, among other things, false science dominates, and there are the next key reasons that contributed to this:

a) Aggression on BiH and war destruction, but also in the wider Balkans, and throughout the surrounding regions, especially in the Middle East (allegedly, local wars are taking place in over 80 countries of the world) led to the destruction of scientific infrastructure (scientific and educational institutions, research centers, laboratories, etc.). Then, the scientific and educational staff was decimated (died, emigrated, stopped in progress and further education), funding of scientific projects was suspended, collapse occurred in all scientific spheres, especially in biomedicine.

b) Financing of scientific and academic institutions in Bosnia and Herzegovina is almost negligible, as well as the financing of potential scientific staff by sending them for education to scientific centers outside Bosnia and Herzegovina and to scientific meetings, where they would present the results of their studies, but also get the knowledge from other researchers. Funds for funding of science in the state sector has been emptied and failed, and the sources of funding of research centers of large companies, such as: Energoinvest, Unioninvest, UNIS, Hidrogradnja, Šipad, whose experts were engaged in large scientific projects together with German, French, English countries which were under the domination of the aforementioned in Africa and the Middle and Far East, brought large revenues to the funds in the former Yugoslavia, and whose funds were enormously used to finance science, scientists and their research, the results of which were applicable in practice and brought profit. Energoinvest's experts constructed computers (IRIS 16 and IRIS 32 and were recognized on the former Yugoslav market and performed equally well with other well-known companies, engaged in bio-robotics, all areas of industry and economy were of a high level and rivaled the strongest European and world companies thanks to the good trained staff in the strongest educational institutions outside BiH. About 2% of the total state budget (GDP) was allocated to science in BiH alone. All of this was almost canceled by the war, and almost all post-war research and projects were, and many are still today financed from EU funds or individual countries of Europe and the world.

c) The new system of education based on the Bologna

concept model, which significantly set back science and education in all areas. In the last 20 years, the author of this article has published over 30 articles related to the Bologna model/concept of education, especially in the field of medicine, he even established his own set of variables for quality assessment of medical education and was the first to introduce web-based medical education in Southeastern Europe, thanks two financed projects in the period 2002-2007. The results showed all the inaccuracies and fallacies of the Bologna model of education, which destroyed the much better Austro-Hungarian model of education in BiH and the countries that were under the administration of Austria since 1878 and rivaled, and in some components even surpassed, the Anglo-Saxon and American model of education. In all scientific spheres, there was a devaluation of educational qualities in every respect, and in particular, better said, the level of both theoretical and practical knowledge declined critically, and the Western European model could not be adequately implemented in Bosnia and Herzegovina because there was neither infrastructure nor sufficient financial resources for it. resources for the implementation of quality educated and knowledgeable educational staff, for the reasons already mentioned in the previous text. Both undergraduate and postgraduate education programs were disrupted, syllabi and teaching content were fragmented, teaching staff were forcibly educated and promoted to teaching positions, which necessitated the proliferation (dysplosion) of textbook writing and the publication of scientific and professional articles necessary for the selection of positions rapid concession and inadequate criteria, and the consequence is "lawyers" with diplomas of deficient and doubtful knowledge for the needs of practical application.

d) Economic and political influence on academic and scientific institutions in Bosnia and Herzegovina, but also in the whole of Southeast Europe, especially influence on universities and academies, elections of bodies and their members and elections to academic positions. The decline in the quality of the educational process at biomedical faculties is still collapsing and there are no solutions for its improvement. Corruption, nepotism, piles of diplomas, plagiarism of the results of other people's research are almost unsolvable, because they are managed by those who have politically reached the positions where they have become decision makers and whose decisions in the institutions of our country and surrounding countries manage everything, including the resources for work and the survival of scientific and academic institutions.

e) Methods, methodology and variables for measuring the scientificity of research results of academic staff and scientists in scientific institutions and research centers are based on insufficiently relevant indicators of scientometric and bibliometric analyses. In the organized systems of other countries, especially when it comes to the evaluation of the results of scientific research through research projects, appropriate scientometric indices are used as indicators of proven scientific knowledge of a researcher, which are reduced to the number of published

works, books, university textbooks, awarding of international and domestic awards etc. Indices, such as Impact and Echo factors, Scopus H index, Google Scholar index, Garfield and Stanford citation lists have dominated the last few years. They are a form of determining one's scientific knowledge - institutional (universities, faculties, academies, etc.) or personal (teaching and scientific staff in the mentioned institutions), but they are insufficiently relevant for the exact verification of one's scientific quality, because they do not realistically show the scientific knowledge and contribution of one institutions or scientists. Scientists whose achievements are world-recognized and have been proven and applicable in practice for decades or centuries are not on those lists at all, because they are more recent, and were established when the mentioned scientists had already died a long time ago. In addition, these lists have so much "distorted" the objectivity of the measurement with variables that unrealistically measure the citation of published works for several reasons - especially emphasizing authorship and co-authorship in published articles, not taking into account the journals and databases where they are deposited (and some journals are discriminated because are not in the databases and have published works that are more often cited in platforms such as ResearchGate or Academia.edu), the contents of the works are not put in the foreground, so for example the addition of authors in the Guidelines, which have become the currently most used form of citation (in which over 1000 co-authors are listed), so that co-authors sometimes have more citations in them than others who are the only author(s) in international or regional and national journals of high scientific rating. In particular, the so-called "familial nepotism" in scientific research (one of the leading problems for editors of journals, especially medical ones, where in the publication of research results, family members are massively added who do not belong to the scientific discipline by vocation or the content of the work in which they did not actually participate.

f) And, finally, all caunted statements in the former text WE NEED to multiplay, as mathematician say, everything with 2×10^3 with consequences of last three years COVID-19 pandemic time- whic spolied our life in all aspects and and still searching solution how to save world population and returned back at least to the condition of work and life style before.

g) All of this can be described and disclosed in greater and more detail and supported by arguments for which more space and time is needed, but it is the "cornerstone" for proving the reason that false science is now more dominant, even globally and not only in countries that have experienced pogrom, such as the Balkan countries and regions where wars have been fought in recent years or are still going on for a long time.

6. CONCLUSION

The false science is a science in which the nonsense served by nutritionists, homeopaths, vaccines, but also cosmetic and pharmaceutical companies is systemati-

cally exposed. Namely, on the one hand, it is said how people misuse science, and on the other hand, how real science works. Alternative medicine methods are very popular in the West. Although some conducted scientific studies have proven that homeopathic remedies are no more effective than a simple placebo, people still happily buy homeopathic preparations. Although the truth should be the aim of scientific research, it is not guiding fact for all scientists. The best way to reach the truth in its study and to avoid the methodological and ethical mistakes is to consistently apply scientific methods and ethical standards in research. When talking about unintentional mistakes, then we think about the discrepancy in terms of methodology of scientific research, while willful misconduct, except deviations from methodological standards in research, involving researchers in dishonesty, or violation of ethical standards (intellectual dishonesty). Some of the errors involving injury have more legal than ethical principles. Scams of the gray zone (undeservedly authorship, multiple publications, manipulation, etc.) are examples of the misuse of science on ethical grounds (un-ethical behaviors are forbidden). Guidelines and rules and their compliance and rules of good scientific practice, are the duties of each research institutions, universities and every individual researchers, no matter what area of science is explored.

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